Gaurang Upadhyay

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Education

Veermata Jijabai Technological Institute, Mumbai **Aug 2024 – Present**
B.Tech in Computer Engineering | CGPA: 8.65

Prakash College of Commerce and Science **Aug 2022 – May 2024**
MHT-CET: 99.98%ile | JEE-Mains: 98.33%ile | HSC: 79.61%

Relevant Coursework

- Data Structures and Algorithms
- Discrete Mathematics
- Object Oriented Programming
- Linear Algebra
- Artificial Intelligence
- Database Management System
- Operating Systems
- Software Engineering
- Design and Analysis of Algorithms

Projects

FinScope | *FastAPI, React, ChromaDB, Scikit-learn, PostgreSQL* **2026**

- Engineered an end-to-end AI-powered financial-statement analysis platform, automating the transformation of unstructured PDF reports and SEC EDGAR 10-K/10-Q filings into structured, queryable data.
- Built a multi-engine document-ingestion pipeline (pdfplumber, PyMuPDF, Tesseract OCR) with LLM-driven field extraction and GAAP/IFRS synonym mapping to normalize varied accounting terminologies.
- Designed an automated ratio engine computing 15+ liquidity, profitability, solvency, and efficiency metrics, paired with a RAG-based chatbot (ChromaDB + Sentence-Transformers) for natural-language queries over uploaded filings.
- Implemented machine-learning-based financial distress and health-risk prediction, side-by-side multi-company comparison views, and secure Google OAuth 2.0/JWT authentication with Supabase PostgreSQL.

FloatChat | *FastAPI, React, Leaflet, ChromaDB, PostgreSQL/SQLite* **2026**

- Developed a full-stack conversational ocean-intelligence platform that translates natural-language queries into real-time insights from global ARGO float oceanographic data.
- Architected a hybrid NL2SQL and RAG engine combining 18 zero-latency SQL templates with a ChromaDB vector store and LLM fallback, safeguarded by an AST-based SQL validator to block unsafe or injected queries.
- Delivered a dual-persona interface (Explorer and Researcher modes) with voice input, and an interactive 360° Leaflet map visualizing 470+ floats worldwide.
- Automated anomaly detection with a rolling z-score model for regional temperature and salinity anomalies, validated through a 49-test pytest suite achieving a 100% pass rate.

Mental Health System | *React, Node.js, Express.js, Supabase (PostgreSQL)* **2025**

- Created a comprehensive full-stack mental health management platform for students, counselors, and administrators using React (Vite), Node.js/Express, and Supabase PostgreSQL.
- Structured 30+ feature modules spanning mood tracking, symptom analysis, sleep and lifestyle tracking, and a sentiment-analyzed journal to build a holistic wellbeing profile for each user.
- Integrated an AI-powered recommendation engine, a conversational chatbot for peer support, and mood-prediction models correlating academic attendance with stress indicators.
- Enforced role-based access control across student, counselor, and admin roles, alongside appointment scheduling, crisis-alert notifications, and a counselor dashboard with weekly analytics and report export.

Technical Skills

Languages: C++, Python, JavaScript, SQL, HTML5, CSS3
Frameworks & Libraries: React, FastAPI, Node.js, Express.js, Tailwind CSS
Databases: PostgreSQL, SQLite, MySQL, ChromaDB
Developer Tools: Git, GitHub, VS Code, Google Colab
Core Concepts: Data Structures & Algorithms, OOP, RAG, Operating Systems, DBMS

Achievements

- Secured **State Rank 67** in Maharashtra in the **MHT-CET** examination.
- **Hacktoberfest 2025 – Open Source Contributor:** Contributed pull requests to open-source GitHub repositories as part of the global Hacktoberfest program, collaborating with maintainers to improve existing projects while gaining hands-on experience with Git workflows and open-source development practices.